LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Pressure altitude, density altitude, geometric, absolute altitude and standard altitude
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Pressure altitude, density altitude, geometric, absolute altitude and standard altitude
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of altitude definitions using IB charts; Instructor supervision required for chart interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the standard sea level pressure?
(b) How does atmospheric pressure change with increasing altitude?
(c) Define the term 'altitude' in general aviation terms.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Geometric altitude definition • Absolute altitude concept • Pressure altitude explanation • Standard altitude criteria • Density altitude derivation	L & Ds	PPT, IB	15 min
(b)	• Relationship between pressure and density • Impact of temperature on density altitude • Calculation of density altitude • Significance of standard atmosphere • Comparison of altitude types	L & Ds	PPT, PDF	20 min
(c)	• Practical application of altitude types • Error sources in altitude measurement • Relevance to aircraft performance • Review of key definitions • Clarification of common misconceptions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between pressure altitude and density altitude.
(b) How does high temperature affect density altitude and aircraft performance?
(c) Why is standard altitude important for flight planning?

12. Home Assignment and Follow Up	Prepare a brief note comparing geometric, absolute, and density altitude, highlighting their practical significance for an Air Agniveer Apprentice.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Static, dynamic and total pressure, principle of A.S.I, Indicated, rectified, equivalent and true air speed, errors in A.S.I.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min